Week 15 Self-Assessment

Correlation

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1 Instructions

This quiz applies everything from Weeks 12-14. You'll practice selecting and interpreting statistical tests, including p-values and effect sizes. This is the last new content before your Research Plan is due.

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2 Part 1: Hypothesis Testing Logic

2.1 Question 1: Null Hypothesis

The null hypothesis (H_0) typically predicts:

- (A) A significant effect
- (B) No effect or no difference
- (C) The direction of the effect
- (D) That the study will be successful

Why do we test the null hypothesis rather than directly testing our experimental hypothesis?

- (A) The null is easier to understand
- (B) We can calculate the probability of our data IF the null were true
- (C) The experimental hypothesis is too complex
- (D) It's just tradition

2.2 Question 2: p-Values

A p-value tells us:

- (A) The probability that the null hypothesis is true
- (B) The probability that our hypothesis is correct
- (C) The probability of obtaining our results (or more extreme) if the null hypothesis were true
- (D) The effect size

If p = .03, this means:

- (A) There's a 3% chance our hypothesis is wrong
- (B) There's a 3% chance of getting these results if there were no real effect

- (C) The effect size is 3%
- (D) 3% of participants showed the effect

2.3 Question 3: Statistical Significance

We typically reject the null hypothesis when:

- (A) $p > .05$
- (B) $p < .05$
- (C) $p = .05$
- (D) p is any value

A result with $p = .08$ is:

- (A) Statistically significant
- (B) Not statistically significant at $\alpha = .05$
- (C) Definitely not a real effect
- (D) A large effect

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3 Part 2: t-Tests Applied

3.1 Question 4: Choosing the Right t-Test

Different participants in each condition →

- (A) Independent t-test
- (B) Paired t-test
- (C) One-sample t-test
- (D) Pearson's r

Same participants measured twice →

- (A) Independent t-test
- (B) Paired t-test
- (C) One-sample t-test
- (D) Pearson's r

3.2 Question 5: DataLab t-Test Selection

Room A vs Room B reaction times (different students in each room):

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) Chi-square

Pre-mood vs Post-mood ratings (same students before and after):

- (A) Independent t-test
- (B) Paired t-test

- (C) Pearson's r
- (D) Chi-square

3.3 Question 6: t-Test Assumptions

Which assumption applies to BOTH independent and paired t-tests?

- (A) Equal group sizes
- (B) Data should be approximately normally distributed
- (C) Participants must be matched on characteristics
- (D) Correlation between variables

The independent t-test additionally assumes:

- (A) Homogeneity of variance (similar SDs in both groups)
- (B) Repeated measures
- (C) Perfect normal distribution
- (D) Large effect size

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4 Part 3: Effect Size

4.1 Question 7: Why Effect Size Matters

Effect size tells us:

- (A) Whether the result is significant
- (B) The p-value
- (C) The magnitude or practical importance of the difference
- (D) The sample size needed

A statistically significant result with a tiny effect size means:

- (A) The finding is very important
- (B) The effect is real but may not be practically meaningful
- (C) We should ignore the result
- (D) The sample was too small

4.2 Question 8: Cohen's d Interpretation

Cohen's d = 0.2 is considered:

- (A) Small
- (B) Medium
- (C) Large
- (D) Very large

Cohen's d = 0.5 is considered:

- (A) Small
- (B) Medium

- (C) Large
- (D) Very large

Cohen's d = 0.8 is considered:

- (A) Small
- (B) Medium
- (C) Large
- (D) Very large

4.3 Question 9: Effect Size Calculation

Cohen's d is calculated as:

- (A) Mean divided by SD
- (B) Difference between means divided by pooled SD
- (C) SD divided by mean
- (D) Sample size divided by variance

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5 Part 4: Interpreting Output

5.1 Question 10: Reading Results

A t-test output shows: $t(48) = 2.35$, $p = .023$, $d = 0.67$

Is this result statistically significant at $\alpha = .05$? TRUE / FALSE

The effect size is:

- (A) Small
- (B) Medium
- (C) Large
- (D) Cannot tell

The degrees of freedom (df) are: __

5.2 Question 11: APA Reporting

The correct APA format for reporting a t-test is:

- (A) t = 2.35, significant
- (B) $t(48) = 2.35$, $p = .023$, $d = 0.67$
- (C) t-test was significant at .05 level
- (D) The groups were different (t = 2.35)

5.3 Question 12: Correlation Reporting

A correlation analysis shows: $r(38) = .45$, $p = .003$

This represents a

- (A) Weak

- (B) Moderate to strong
- (C) Perfect
- (D) Non-significant

positive relationship.

The result IS statistically significant. TRUE / FALSE

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6 Part 5: Complete Test Selection

6.1 Question 13: Design to Test Matching

Fill in the correct test for each scenario:

Comparing **two groups of different people**: ____________________

Comparing the **same people at two time points**: ________________

Examining the **relationship between two continuous variables**: _______________

6.2 Question 14: Non-Parametric Alternatives

When data violate normality assumptions:

Independent t-test →

- (A) Mann-Whitney U
- (B) Wilcoxon signed-rank
- (C) Spearman's rho
- (D) Chi-square

Paired t-test →

- (A) Mann-Whitney U
- (B) Wilcoxon signed-rank
- (C) Spearman's rho
- (D) Chi-square

Pearson's r →

- (A) Mann-Whitney U
- (B) Wilcoxon signed-rank
- (C) Spearman's rho
- (D) Chi-square

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7 Part 6: MCQ Exam Practice

7.1 Investigation 1

A researcher tests whether a mindfulness intervention reduces stress. 30 participants are randomly assigned to either a mindfulness group or a control group. Stress is measured after 4 weeks.

1. **Design**:

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

2. **IV**:

- (A) Group (mindfulness vs control)
- (B) Stress level
- (C) Time (4 weeks)
- (D) No IV

3. **DV**:

- (A) Group assignment
- (B) Stress level
- (C) Mindfulness practice
- (D) Participant age

4. **Appropriate test**:

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) Mann-Whitney U

5. **Results: $t(28) = 2.89$, $p = .007$, $d = 1.08$. This is**:
   - Significant? TRUE / FALSE
   - Effect size:

- (A) Small
- (B) Medium
- (C) Large
- (D) Cannot determine

6. **The correct conclusion is**:

- (A) The mindfulness group had significantly lower stress than controls
- (B) There was no significant difference
- (C) The control group had lower stress
- (D) We cannot draw any conclusions

7.2 Investigation 2

A researcher measures students' confidence before and after receiving exam feedback.

1. **Design**:

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

2. **IV**:

- (A) Time (before vs after feedback)
- (B) Confidence level
- (C) Exam grade
- (D) No IV

3. **DV**:

- (A) Time point
- (B) Confidence level
- (C) Feedback content
- (D) Student ID

4. **Appropriate test**:

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) Wilcoxon signed-rank

5. **Results: $t(24) = -3.45$, $p = .002$, $d = 0.69$. The negative t-value means**:

- (A) The result is not significant
- (B) Confidence decreased (before > after)
- (C) The calculation is wrong
- (D) The effect size is negative

7.3 Investigation 3

A researcher examines whether study hours predict exam scores in 60 students.

1. **Design**:

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

2. **IV**:

- (A) Study hours

- (B) Exam scores
- (C) Number of students
- (D) No IV

3. **Appropriate test**:

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) ANOVA

4. **Results: $r(58) = .52$, $p < .001$. This represents**:

- (A) A weak relationship
- (B) A moderate to strong positive relationship
- (C) No relationship
- (D) A negative relationship

5. **Can we conclude studying CAUSES better grades?** TRUE / FALSE

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8 Summary Checklist

Before moving on, make sure you can:

- ☐ Explain the logic of hypothesis testing (null hypothesis, p-value)
- ☐ Interpret p-values correctly (NOT the probability the null is true)
- ☐ Select the appropriate test for any design (independent t, paired t, Pearson's r)
- ☐ Identify non-parametric alternatives
- ☐ Interpret and report effect sizes (Cohen's d, r)
- ☐ Read and interpret statistical output
- ☐ Report results in APA format

> ! Research Plan Due Next Week
>
> Your Research Plan is due Week 16. You need to specify your hypothesis, design, IV, DV, and statistical test. This quiz covers all the content you need!